

Getting The Right Driver For Your LED Application

Semiconductor industry is moving from simply selling power supply regulators as LED drivers to making dedicated products that can drive LEDs in various applications. Selection of the LED driver and specifications depends on the kind of application that you are designing

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Light-emitting diodes (LEDs) emit light when current passes through them. And to provide a constant current to an LED, you need a source—an LED driver. Earlier, when LEDs were used just as an indicator, a constant DC source and a resistor were sufficient to light them. As LED applications evolved from basic ‘indicator’ to a variety of high-current lighting applications, the drivers too evolved in sync leading to a new semiconductor innovation.

There are a variety of LED drivers available today having different fea-

tures for different applications. Senith Srinivasan, director and head of industrial and multi-market division, Infineon, says, “Every LED application has distinct power requirements, features and level of protection. You find linear type of LED drivers, switching LED drivers and multiple-channel LED drivers with current rating to fit the target applications in the market.”

Teo Chee Keong, Panasonic Semiconductor Sales Asia (PSCSA), shares “Higher efficiency, higher accuracy, multiple protection functions, minimal component count, smart dimming using high-performance pulse-width

modulation (PWM) control and flexibility in design are the trends in today’s LED drivers.” All these trends have led to a variety of offerings in LED drivers.

LED applications: what’s hot

The demand for LEDs in applications ranging from display backlight units to LED luminaires has been continuously on the rise for the last few years. One of the reports by Petrov Group, titled *LED Driver ICs for Lighting and Display*, says, “The market for LED driver ICs is the only near- and long-term growth market in the semiconductor industry featuring a ‘greater than 20 per cent’ average CAGR.”

Higher efficiency and lower pricing have remarkably increased the use of LEDs in display backlight units of notebook PCs, medium- and large-size LCD TVs, and full-colour signages.

Keong says, “The displays are either made of LEDs or are LCDs which need backlight provided by LEDs.”

A lot of innovative design is happening in the lighting segment, with LEDs being used for indoor as well as outdoor lighting. Some of the indoor lighting solutions include portable emergency LED lighting equipment (lanterns and torches), LED bulbs, LED tubes, LED spotlights and LED down-lights. For 100V-230V AC and small-form-factor retrofit lamps (MR16, GU10, etc), companies have started providing unique solutions that provide protection to the driver



Maxim two-string HB LED driver for automotive applications